

## CS 270 – Lab 9 Q2

Question 2: (H a b)

```

;input-contract: a and b are nonnegative integers
;output-contract: (H a b) is a recursively defined integer
;
; -- pattern redacted for students
(define (H a b)
  (if (< a b) 0 (+ 1 (H (- a b) b))))

```

Use Equational Reasoning to prove (H 3 2) evaluates to 1.

| Expression                               | Justification                          |
|------------------------------------------|----------------------------------------|
| (H 3 2)                                  | Premise                                |
| (if (< a b) 0 (+ 1 (H (- a b) b))))      | Apply definition of H with a=3 and b=2 |
| (if #f 0 (+ 1 (H (- 3 2) 2)))            | Evaluate <                             |
| (+ 1 (H (- 3 2) 2))                      | Evaluate if                            |
| (+ 1 (H 1 2))                            | Evaluate -                             |
| (+ 1 (if (< 1 2) 0 (+ 1 (H (- 1 2) 2)))) | Apply definition of H with a=1 and b=2 |
| (+ 1 (if #t 0 (+ 1 (H (- 1 2) 2))))      | Evaluate < (less than)                 |
| (+ 1 0)                                  | Evaluate if                            |
| 1                                        | Evaluate +                             |